
ROTATIONAL FREQUENCY SHIFTS

In the late 1970s Garetz and Arnold demonstrated that when a circularly polarised ($\sigma = \pm 1$) wave was transmitted through a half-wave plate rotating at $\Omega/2$, its frequency was shifted by $\sigma\Omega$ (6.1). Garetz recognised that the frequency shift was an example of a more general angular Doppler shift (6.2, **Paper 6.1**), which was manifest whenever light was emitted, absorbed or scattered by rotating bodies. The frequency shift can be understood in terms of energy conservation or, more sophisticatedly, by a Jones matrix analysis. Subsequently, the same type of experiment was described in terms of a dynamically evolving Berry phase (6.3). The shift is analogous to the speeding up or slowing down of a clock hand when the clock is placed on a rotating turntable.

In 1993 van Enk argued that when LG modes were transformed to HG modes where $n + m = 1$ and back again, there should be a geometric phase equivalent to the Pancharatnam phase obtained for polarisation (6.4). The way in which astigmatic optical elements transformed the phase of helical phasefronts, analogous to the way birefringent components transformed the phase of polarised waves, had already been demonstrated by Beijersbergen *et al.* (6.5, **Paper 3.5**). Nienhuis predicted that the rotation of a cylindrical-lens π -mode converter would impart a frequency shift to the transmitted LG mode (6.6, **Paper 6.2**). For a mode converter rotation frequency of $\Omega/2$, the predicted frequency shift of the light beam should be $l\Omega$. The waveplate and the mode converter perform a mirror inversion of the electric field and phase structure respectively. It follows that a rotation of the optical component by α rotates the polarisation, or phase structure, by 2α and so a $\Omega/2$ rotation frequency of the optical component rotates the beam at Ω . In each case, the frequency shift is equal to the rotation rate of the beam multiplied by the angular momentum per photon. Both shifts are readily explicable in terms of transformations on the Poincaré sphere or by Jones matrices or by their equivalent for helically phased modes (6.7, **Paper 6.3** and 6.8, **Paper 2.4**).

In 1998, Courtial *et al.* (6.9) generated a variety of LG modes in the millimetre-wave region of the spectrum and passed them through a rotating π -mode converter based on a Dove prism: conveniently, this device does not rotate the polarisation state (6.10). They confirmed the $l\Omega$ frequency shift and explained it both in terms of earlier work, a quantum mechanical rotational frequency shift identified by Bialynicki-Birula and Bialynicka-Birula (6.11, **Paper 6.4**) and an azimuthal Doppler shift applied to atomic transitions (6.12, **Paper 6.5**). This frequency shift is simply explicable in terms of the electric field. Whereas the phasefront of a planar wave is invariant under rotation, the phasefront of a helical beam is not. For helical phase fronts, a rotation of the beam is

indistinguishable from its temporal evolution and one rotation of the beam introduces a phase change of l cycles. A beam rotation frequency of Ω thus produces a frequency shift of $l\Omega$.

Courtial *et al.* (6.13, **Paper 6.6**) also experimented with circularly polarised LG modes, rotating the whole beam by means of a Dove prism and half-waveplate combined. They confirmed that the result was not two individual frequency components relating to the shifts arising from the spin and orbital angular momentum, but a single frequency shift that was proportional to the total angular momentum, that is $(l + \sigma)\Omega$, as predicted by the Birulas (6.11). This equivalence of action between spin and orbital angular momentum is by no means general. Again the effect can be understood in terms of the electric field distribution. A cross-section through the beam shows that the effect of circular polarisation and the helical structure is to produce an $(l + \sigma)$ -fold rotational symmetry, such that upon one rotation the phase change is $(l + \sigma)$ cycles.

This general form of the angular Doppler shift should not be confused with the Doppler shift associated with rotating galaxies and other bodies. The latter effect is simply a manifestation of the translation Doppler effect where the rotation of an object results in a component of velocity in the line of sight. The angular Doppler shift is a maximum in the direction of the angular momentum vector where the linear Doppler shift is zero. Whereas the translation Doppler shift can be expressed in terms of the linear momentum multiplied by the linear velocity per photon, $k_z v_z$, the angular or rotational Doppler effect is the angular momentum per photon multiplied by the annular velocity, that is $[k_\phi(r)/r][rv_\phi]$, or $k_\phi(r)v_\phi$.

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Angular Doppler effect

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An angular analog of the Doppler effect arising from the quantum of angular momentum carried by circularly polarized photons is presented and developed. Applications to rotational Raman scattering, fluorescence doublets, controlled frequency shifting of light, rotation-induced optical activity, and the measurement of rotational motion of small particles are discussed.

INTRODUCTION

That a photon carries a quantum of angular momentum has some interesting consequences concerning the way it interacts with matter. It is responsible for frequency shifts in circularly polarized light interacting with bodies in angular motion. Early mention of frequency shifts that are due to light interacting with rotating bodies dates back to Airy's conception of revolving plane-polarized light as the superposition of right and left circularly polarized components of different frequencies and Righi's experimental tests of this idea using a rotating Nicol.¹

In the course of this Letter, we show that phenomena as diverse as rotational Raman spectra,² fluorescence doublets, and certain frequency-shifting devices^{3,4} are manifestations of the angular Doppler effect (ADE). Other applications include the measurement of small-particle rotational motion and rotation-induced optical activity.

We are concerned with angular motion with an axis of rotation that is parallel to the direction of propagation. Such angular motion is not to be confused with angular motion with an axis of rotation that is normal to the propagation vector; the latter is responsible for frequency shifts that are fully described by the linear Doppler effect.⁵ We are also not concerned with the relativistic transverse Doppler effect, which is due to linear motion perpendicular to the propagation vector.⁶

Although quantum mechanics is used as a convenient device for simplifying the discussion of the ADE, the ADE is essentially a classical effect.

1. ANGULAR MOMENTUM OF LIGHT

Circularly polarized (CP) light is of particular interest because it is that electromagnetic radiation that is an eigenstate of the angular momentum operator L_z , with eigenvalues $\pm\hbar = \pm h/2\pi$, where h is Planck's constant, corresponding to right and left circular polarization (RCP and LCP, respectively). This connection between circular polarization and angular momentum predates quantum theory. Poynting first suggested that circularly polarized electromagnetic waves should carry angular momentum of magnitude $\lambda/2\pi$ times the linear momentum of the wave,⁷ where λ is the wavelength. This is also the quantal result since the linear momentum p has the

value $h/\lambda = h\nu/c$, where ν is the frequency and c is the speed of light. Thus a photon carries linear momentum proportional to ν/c , whereas its angular momentum is independent of frequency. These facts dictate some important differences in the details of the linear and angular Doppler effects.

2. ANGULAR DOPPLER EFFECT

The linear Doppler effect has two aspects: (1) that light emitted from translating bodies suffers a shift $\Delta\nu = \nu v/c$, where v is the relative velocity between observer and body; and (2) that light interacting with (specifically, reflected from) translating bodies suffers a shift $\Delta\nu = 2\nu v/c$.

The angular Doppler effect displays two analogous aspects, which we treat separately.

A. Light Emitted from Rotating Bodies

Our development of the ADE closely parallels standard derivations of the linear Doppler effect, as given by Sommerfeld and others. A full treatment requires relativity. Since $v \ll c$ for the phenomena with which we are concerned, computations can be carried out classically. We follow a corpuscular description similar to one used by Sommerfeld.⁸

Consider a particle rotating with angular frequency ν_{rot} . If a CP photon is emitted normal to the plane of rotation, conservation of angular momentum requires that $\Delta L = \pm\hbar$. (Whether the sign is plus or minus depends on the relative senses of the circular polarization of the light and the rotational motion of the particle.) Since the rotational kinetic energy is given by $E = L^2/2I$, where I is the moment of inertia of the particle, conservation of energy dictates that

$$\frac{L_1^2}{2I} + E_1 = \frac{L_2^2}{2I} + E_2 + h(\nu + \Delta\nu), \quad (1)$$

where L_1 and L_2 are particle angular momenta before and after emission, respectively, E_1 and E_2 represent the internal electronic energies of the emitting atom or molecule before and after emission, respectively, ν is the frequency of the emission when the particle is at rest, and $\Delta\nu$ is the angular Doppler shift in the frequency that is due to particle rotation.

The rotating particle may be some macroscopic body on which an emitting molecule or light source is fixed, or it may

be the emitting molecule itself. In the latter case, $L^2/2I$ is the nuclear-rotational contribution to the molecular kinetic energy. Interpreting Eq. (1) in terms of a single molecule, we are thus assuming that the electronic Hamiltonian is separable from the nuclear-rotational Hamiltonian and that the rotational motion may be treated classically. The first assumption is essentially the Born-Oppenheimer approximation⁹; the second is appropriate in the limit that $L \gg \hbar$, or equivalently, that the rotational quantum number $J \gg 1$. This situation is frequently encountered with gas-phase molecules of moderate molecular weight at room temperature. [For iodine (I_2) at room temperature, the most probable J is approximately 70.] The practice of treating some degrees of freedom classically and others quantum mechanically is not a new one; the translational energy of atoms and molecules is commonly treated classically. We have simply shifted nuclear rotation into this classical realm.

Realizing that $E_1 - E_2 = h\nu$ yields

$$h\Delta\nu = \frac{L_1^2 - L_2^2}{2I} = \frac{(L_1 - L_2)(L_1 + L_2)}{2I} = \frac{\Delta L}{I} \left(\frac{L_1 + L_2}{2} \right), \quad (2)$$

which is $\approx (L\Delta L/I)$ to first order in L . However, $\Delta L = \pm\hbar$ and $L = I\omega_{\text{rot}} = 2\pi I\nu_{\text{rot}}$, where ω_{rot} is the particle's angular velocity in rad/sec, so that

$$h\Delta\nu = \pm 2\pi\hbar\nu_{\text{rot}} = \pm h\nu_{\text{rot}} \text{ or } \Delta\nu = \pm\nu_{\text{rot}}. \quad (3)$$

We have assumed here that, in the rotating molecule case, the molecular electronic angular momentum remains unchanged on emission. Thus we are limiting that portion of our treatment to transitions of the sort $\Sigma \rightarrow \Sigma$ in diatomic molecules.

An easy way to visualize the shift just described is to consider that CP light of frequency ν is represented by a rotating electric field vector, which makes ν rev/sec. An observer in a reference frame rotating at angular frequency ν_{rot} would count $(\nu - \nu_{\text{rot}})$ or $(\nu + \nu_{\text{rot}})$ revolutions in 1 sec, depending on whether the electric field rotation was in the same or the opposite sense as the frame rotation, respectively.

B. Light Interacting with Rotating Bodies

The next step is to determine the angular analog of light reflected from a translating mirror. A mirror reverses the sign of the linear momentum vector of the light. The angular equivalent is an interaction in which the sign of the angular momentum vector is reversed, i.e., in which RCP is converted to LCP or vice versa. A half-wave retardation plate serves this function.³ The change in angular momentum of a CP beam suggests a torque applied to the wave plate. Such torques have been experimentally measured by Beth.¹⁰ If the wave plate is rotating, this torque is applied through an angular displacement, meaning that work is done on or by the wave plate:

$$W = \int \tau d\theta, \quad (4)$$

where W is the work, τ is the applied torque, and θ is the angular displacement.

For a single CP photon, the change in angular momentum $\Delta L = \pm 2\hbar$ (cf. the emissive case, where $\Delta L = \pm\hbar$). For a wave plate rotating at angular velocity ω_{rot} , $L = I\omega_{\text{rot}}$ and $E_{\text{rot}} = L^2/2I$. To first order, the change in energy of the plate is

Table 1. Comparison of Linear and Angular Doppler Frequency Shifts

	Emissive Case	Interactive Case
Linear Doppler	$\pm\nu_0(v/c)$	$\pm 2\nu_0(v/c)$
Angular Doppler	$\pm\nu_{\text{rot}}$	$\pm 2\nu_{\text{rot}}$

$$\Delta E = \frac{L\Delta L}{I} = \pm 2\hbar\omega_{\text{rot}} = \pm 2h\nu_{\text{rot}}. \quad (5)$$

To conserve energy, this requires the photon to gain or lose this energy, which is manifested in a frequency change

$$\Delta\nu = \pm 2\nu_{\text{rot}}, \quad (6)$$

i.e., the frequency of an incident CP beam whose electric field is rotating in the same sense as the wave plate is downshifted by $2\nu_{\text{rot}}$, whereas a CP beam of the opposite sense is upshifted by $2\nu_{\text{rot}}$. This is analogous to the linear momentum transfer that occurs in the linear Doppler effect, which causes a frequency shift $\Delta\nu = \pm 2\nu_0 v/c$.

Interactions involving the scattering of light from rotating bodies can yield an unshifted component to the scattered light, in addition to the two shifted components described above.² This arises from photons scattered with their angular momentum vectors unchanged. In this case, $\Delta L = 0$ and $\Delta\nu = 0$.

Table 1 summarizes the frequency shifts associated with the linear and angular Doppler effects for both emissive and interactive cases. The factor of 2 in the interactive cases arises because the linear or angular momentum vector is reversed, causing a change in momentum of *twice* the magnitude of the momentum of the photon. The factor ν_0 for the linear Doppler effect arises from the fact that the photon linear momentum is proportional to the frequency of the light, whereas the angular momentum is independent of frequency.

3. APPLICATIONS

A. Rotational Structure of Emission Spectra: Fluorescence Doublets

In terms of the ADE, the rotational structure of vibronic transitions becomes illuminated. Consider the $B \rightarrow X$ emission of the diatomic molecule, iodine. A molecule in rotational quantum level J rotates with $L = \hbar J = I\omega$ or with a classical frequency $\nu_{\text{rot}} = \pm hJ/4\pi^2 I$. If the I_2 molecule at rest would emit light at ν_0 , the ADE would predict RCP and LCP emission doublets observable at $\nu = \nu_0 \pm hJ/4\pi^2 I$. This is observed quantum mechanically as a result of the selection rule $\Delta J = \pm 1$, which, in the classical limit (large J), yields the same result. This is obviously no coincidence, since the rotational selection rule is simply a consequence of conservation of angular momentum coupled with the fact that photons carry a quantum of angular momentum. In a sense, the existence of fluorescence doublets is a classical phenomenon.

B. Rotational Raman Scattering

Rotational Raman scattering represents a case of interactive ADE. As such, the scattering of light of frequency ν_0 from a rotating diatomic molecule, such as I_2 in rotational level J (classical rotational frequency of $\nu_{\text{rot}} = hJ/4\pi^2 I$), yields

scattered components at frequencies ν_0 and $\nu_0 \pm 2hJ/4\pi^2I$. Because of quantum mechanical selection rules $\Delta J = 0, \pm 2$, this same result is obtained quantum mechanically in the classical limit. Subtle differences in the two cases are discussed by Newburgh and Borgiotti.²

C. Variable Frequency Shifting of CP Light

The ADE is the basis for a device capable of shifting the frequency of a CP beam of light by an arbitrary amount.³ It involves sending CP light through a rotating quartz half-wave plate. The resulting light displays a frequency shift of $\pm 2\nu_{\text{rot}}$, where ν_{rot} is the rotational frequency of the wave plate. Because of physical limitations in rotating macroscopic quartz plates without distortion, achievable shifts are limited to the kilohertz regime, although megahertz shifts are possible if the retardation is carried out electro optically.

D. Other Applications

As mentioned at the end of Section 2.A, incident RCP and LCP waves of the same frequency are observed to have different frequencies in a rotating reference frame. If the rotating medium exhibits dispersion in absorption or refractive index, an apparent optical activity should be observed, since the right and left CP components are absorbed or refracted by different amounts by the medium. Such induced circular dichroism should be substantial in the vicinity of sharp absorption peaks or band edges.

The ADE can be used to measure the rotational velocity of small particles through the detection of frequency-shifted components of CP light scattered from such particles, the sign of the shift yielding the sense of the particle rotation. Finally, shifted frequency components in light scattering measurements in liquids that are due to processes such as rotational

diffusion and rotational Brownian motion can be interpreted in terms of the ADE.

In summary, the angular Doppler effect is a novel optical principle that characterizes frequency shifts in CP light, which is emitted from or interacts with rotating bodies. The principle arises from simple conservation of energy and angular momentum considerations and has application to a variety of spectroscopic, light scattering, and optical processing phenomena.

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Doppler effect induced by rotating lenses

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Abstract

The orbital angular momentum inherent to light beams with a helical wavefront can be transferred to non-isotropic lenses. A system of three cylindrical lenses suffices to transform an arbitrary paraxial input beam by inverting one transverse direction of the mode function. This also inverts the orbital angular momentum of a Laguerre–Gaussian beam with mode index m . The corresponding transfer of angular momentum shows up directly as a frequency shift $2m\Omega$ when the lens system is set in rotation at frequency Ω around the axis.

1. Introduction

An optical element that rotates about its axis at a uniform angular velocity Ω will usually generate sidebands in the spectrum of a light beam. When the element affects only the light polarization, and the monochromatic input beam has frequency ω , the sidebands can only occur at the frequencies $\omega \pm 2\Omega$. In special cases, the frequency is simply shifted by 2Ω . An example is found in Ref. [1], where the observed frequency shift of a linearly polarized beam after a double pass through a rotating quarter-wave plate has been interpreted as a time-dependent manifestation of a geometric phase. The closed trajectory on the Poincaré sphere describing the polarization encloses a solid angle that varies linearly with time, and the resulting linear variation of the Pancharatnam phase corresponds to a frequency shift. From a less fancy point of view, frequency shifts due to a rotating birefringent plate may also be regarded as resulting from the energy exchange between the light beam and the plate. The origin of the work that the light exerts on the rotating plate lies in the torque that accompanies the exchange of angular momen-

tum, for instance when a half-wave plate inverts the handedness of circular polarization [2]. In this picture, the frequency change has a dynamical rather than a geometrical nature. The mechanical action of this torque has been demonstrated in an early experiment by Beth [3]. The physical origin of the geometric Pancharatnam phase in transfer of angular momentum has been suggested before [4,5].

In addition to the angular momentum inherent to circular polarization, a light beam can also have an angular momentum along the axis that results from the helical nature of the wavefronts [6]. This orbital angular momentum results from the density of transverse momentum, which in turn corresponds to the transverse phase gradients that generally exist in a beam of finite extent. Therefore, one may expect that sidebands generated by rotating lenses can likewise be viewed as resulting from the exchange of orbital angular momentum between beam and lens. This idea will be explored in the present paper. Whereas the polarization-dependent angular momentum is limited to a single unit $\pm\hbar$ per photon, the orbital angular momentum can in principle amount to a large number of units. As a result, one predicts that

the frequency change can be many times the rotation frequency Ω . In some cases, a cyclic transformation of the state of the orbital angular momentum of a paraxial beam can be accompanied by a geometric phase change [7]. A frequency shift will then arise when the cyclic transformation is performed by rotating optical elements, so that the geometric phase varies linearly with time. We describe an inverter consisting of three astigmatic lenses, which transforms an arbitrary paraxial beam with orbital angular momentum $\hbar m$ per photon into a beam with the opposite value $-\hbar m$. In contrast to other mode converters, the present lens system need not be adapted to the specific properties of the input beam, such as its Rayleigh range or the location of its focus. Then the frequency shift $2m\Omega$ induced by rotation of the lenses can be directly interpreted as arising from the torque on the lenses that accompanies the transfer of orbital angular momentum.

2. State-vector description of time-dependent paraxial beams

2.1. Free propagation

A moving optical element, such as a receding mirror or a rotating lens, will in general introduce non-monochromaticity into a light beam. Therefore, we use a time-dependent version of the paraxial wave equation. A light beam propagating in free space is described by Maxwell's wave equation for the electric field,

$$\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (1)$$

For a nearly monochromatic beam with polarization $\boldsymbol{\varepsilon}$ and propagating along the z -axis the electric field can be factorized as

$$\mathbf{E}(\mathbf{r}, t) = \boldsymbol{\varepsilon} u(\mathbf{r}, t) e^{ikz - i\omega t}, \quad (2)$$

where the field amplitude u obeys the time-dependent paraxial wave equation

$$\left(\frac{\partial}{\partial z} + \frac{1}{c} \frac{\partial}{\partial t} \right) u = \frac{i}{2k} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u. \quad (3)$$

This equation results after substituting Eq. (2) in the wave equation (1), while neglecting second derivatives of u with respect to z and t . This is justified when the amplitude u varies little with z and t over a wavelength or an optical cycle. Eq. (3) generalizes the standard paraxial wave equation for monochromatic beams, in which u is independent of time [8]. Then the time derivative in Eq. (3) may be omitted, and the resulting equation has the same form as Schrödinger's equation for a free particle in two dimensions, where the z coordinate plays the role of time [9]. This suggests describing the amplitude of a monochromatic paraxial beam for a given value of z in terms of a state vector $|u(z)\rangle$, in analogy to a quantum-mechanical time-dependent state vector $|\psi(t)\rangle$ [10]. The evolution of a quantum-mechanical state as described in terms of linear operators then corresponds to the propagation of the light beam. In the present case of non-monochromatic paraxial beams, the state vector depends both on z and t , and the propagation equation takes the linear form

$$\begin{aligned} & \left(\frac{\partial}{\partial z} + \frac{1}{c} \frac{\partial}{\partial t} \right) |u(z, t)\rangle \\ &= -\frac{i}{2k} \left(\hat{P}_x^2 + \hat{P}_y^2 \right) |u(z, t)\rangle, \end{aligned} \quad (4)$$

where we introduced the components of a momentum operator

$$\hat{P}_x = -i \frac{\partial}{\partial x}, \quad \hat{P}_y = -i \frac{\partial}{\partial y}. \quad (5)$$

The free propagation of a beam over a distance Z can be expressed in terms of a propagation operator

$$\hat{U}(Z) = \exp \left(-\frac{iZ}{2k} \left(\hat{P}_x^2 + \hat{P}_y^2 \right) \right), \quad (6)$$

in analogy to the quantum-mechanical evolution operator. However, the effect of free propagation constitutes a slight generalization with respect to quantummechanical evolution, in that the effect of retardation must be included. The solution of Eq. (4) obeys the equality

$$|u(z, t)\rangle = \hat{U}(Z) \left| u \left(z_0, t - \frac{Z}{c} \right) \right\rangle, \quad (7)$$

with $Z = z - z_0$ the traversed distance. Eq. (7) relates the field amplitude at the position z at time t to the amplitude at position z_0 at an earlier time. The time difference is simply the travel time of light between the two positions.

2.2. Ideal lenses

The effect of a thin dielectric object placed in the beam is described by a phase change that depends on x and y . The layer must be so thin that transverse propagation in the layer is negligible, so that for each value of x and y the output amplitude is equal to the input amplitude multiplied by the phase factor $\exp(i\eta)$ with

$$\eta(x, y) = k(n - 1)a(x, y). \quad (8)$$

Here n is the refractive index of the dielectric, and a is its thickness. A purely cylindrical lens with axis oriented at an angle α with respect to the x -axis can then be described by the operator $\hat{S}$, that is equal to [10,11]

$$\hat{S}(\alpha) = \exp\left(-\frac{ik}{2f}(\hat{X} \cos \alpha + \hat{Y} \sin \alpha)^2\right), \quad (9)$$

where f is the focal length, and where by definition the operators $\hat{X}$ and $\hat{Y}$ multiply the amplitude function $u(x, y)$ with x or y . The orientation dependence of the lens operator is conveniently expressed as

$$\hat{S}(\alpha) = \hat{R}(\alpha) \hat{S}(0) \hat{R}^\dagger(\alpha), \quad (10)$$

i.e. in terms of the rotation operator

$$\hat{R}(\alpha) = \exp(-i\alpha \hat{J}), \quad (11)$$

with

$$\hat{J} = \hat{X}\hat{P}_y - \hat{Y}\hat{P}_x = -i\frac{\partial}{\partial\phi} \quad (12)$$

the angular momentum operator. When the lens is rotating at a constant rotation frequency Ω , the operator $\hat{S}$ attains a time dependence imposed by the time-dependent angle $\alpha = \Omega t$. For an arbitrary incident monochromatic beam, the output amplitude is obtained simply by multiplying the input amplitude by a phase factor. For the lens indicated by Eq. (9), the factor can be expressed in polar coordinates as

$$\exp\left[-\frac{ik}{2f}\rho^2 \cos^2(\Omega t - \phi)\right]$$

$$= \exp\left(-\frac{ik\rho^2}{4f}\right) \sum_{n=-\infty}^{\infty} J_n\left(\frac{k\rho^2}{4f}\right) \times \exp[2in(\Omega t - \phi - \pi/4)], \quad (13)$$

with J_n the ordinary Bessel functions. This demonstrates that for a monochromatic input beam, the output of the rotating lens is highly non-monochromatic. It contains sidebands at all frequencies $\omega + 2n\Omega$.

When a time-dependent input beam passes a rotating lens at the position z_0 , the output state vector at position z is can be expressed as

$$|u_{\text{out}}(z, t)\rangle = \hat{U}(Z) \hat{S}(t - Z/c) |u_{\text{in}}(z_0, t - Z/c)\rangle, \quad (14)$$

with $Z = z - z_0$. This is easily verified by noting that the output beam obeys the time-dependent paraxial equation (4), while at $z = z_0$ it is correctly related to the input by the time-dependent lens operator.

3. Conversion of paraxial beams

3.1. HG-LG conversion

A Hermite-Gaussian (HG) beam can be converted into a Laguerre-Gaussian (LG) beam by using two cylindrical lenses. This has been demonstrated by Tamm and Weiss for the lowest-order Hermite-Gaussian mode [12]. In fact, the technique can be applied to modes of any order [13]. The setting of the pair of lenses must be such that they impose a difference equal to $\pi/2$ in the Gouy phase of the x -dependence and the y -dependence of the mode function. This can be done by orientating the axis of two identical cylindrical lenses by an angle of 45° with respect to the symmetry axis of the incident HG mode. The two lenses must be symmetrically positioned at opposite sides of the focus, at a distance

$$D = \beta \tan \frac{\pi}{8} = \beta(-1 + \sqrt{2}) \quad (15)$$

from the focus, where β is the Rayleigh range of the input beam, which is half its confocal parameter. The focal length of the cylindrical lenses must be equal to

$$f = D\sqrt{2} = \beta(2 - \sqrt{2}). \quad (16)$$

The Rayleigh range β' of the transformed beam factor in the astigmatic region between the two lenses is then equal to $D^2/\beta = \beta(3 - 2\sqrt{2})$.

3.2. Harmonic-oscillator picture

It has been shown [14] that the connection between various higher-order Gaussian modes is mathematically identical to the eigenstates of a two-dimensional quantum harmonic oscillator, and that the change of the Gouy phase during propagation has the same effect as the corresponding phase evolution of the two components of the harmonic oscillator. Higher-order modes can always be described as arising from the action of a raising operator on the fundamental mode. This fundamental mode has a Gaussian shape, and for each value of z it is fully determined by the radius of curvature s of the wavefronts and by the spot size γ . For an astigmatic beam, these parameters are different for two orthogonal transverse directions. The local values of s and γ determine both the Rayleigh range β and the distance z from the focus, by the relation

$$\frac{1}{\gamma^2} - \frac{ik}{s} = \frac{k}{\beta + iz}. \quad (17)$$

This equation determines the z -dependence of γ and s during free propagation. The corresponding change of the Gouy phase χ can be expressed by the differential

$$d \tan \chi = dz/\beta. \quad (18)$$

A thin lens with focal length f does not affect the spot size γ , but it affects the curvature radius s , according to the identity

$$\frac{1}{s_+} = \frac{1}{s_-} - \frac{1}{f}. \quad (19)$$

Of course, a cylindrical lens only affects the curvature radius in the plane in which the lens is focusing. An attractive feature of the harmonic-oscillator analogy is that the effect of propagation and of ideal lenses on a Gaussian beam of arbitrary order can be expressed solely in terms of the beam parameters γ and s , which determine the local Gaussian envelope of the mode function.

A HG beam is analogous to a factorized state of the harmonic oscillator, with a well-defined excita-

tion number for the oscillation in two orthogonal directions. An LG beam corresponds to a state of the two-dimensional oscillator that is an eigenstate of both energy and angular momentum. When the azimuthal mode number of the LG beam is equal to m , corresponding to the dependence $\sim \exp(im\phi)$ of the mode function u , the beam carries an orbital angular momentum in the propagation direction that is equal to $\hbar m$ per photon [6]. The conversion of a HG into an LG beam is analogous to the transformation of a factorized oscillator state with symmetry axis rotated over 45° with respect to the x -axis to an angular momentum eigenstate, simply by imposing a $\pi/2$ phase difference between the oscillation in the two orthogonal directions. The resulting angular momentum quantum number m is then equal to the difference of the two one-dimensional excitation numbers.

3.3. LG inversion

From the harmonic-oscillator analogy it follows that an LG beam $\sim \exp(im\phi)$ is converted to an LG beam $\sim \exp(-im\phi)$ by a lens system that imposes a phase difference π for the x dependence compared with the y dependence, in the same way as a harmonic oscillator eigenstate of angular momentum is transformed into the state with opposite eigenvalue by imposing a phase difference π for the oscillation in two orthogonal directions. It is impossible to make such a converter between two opposite LG beams with just two cylindrical lenses [11]. The reason is basically that the phase difference can only build up in the region between the two lenses, where the beam is astigmatic. Since the phase change of a Gaussian beam over a focus is asymptotically equal to π , this difference of two phase changes is necessarily smaller than π .

On the other hand, one would expect that such an LG inverter can be constructed by a combination of three parallel astigmatic lenses. To be specific, we consider a symmetric configuration, with three parallel cylindrical lenses that are focusing in the xz -plane, while leaving the y dependence of the mode function unaffected. Two identical cylindrical lenses are located at the positions $z = \pm D$, and a third cylindrical lens is positioned at $z = 0$. The two outer lenses have a focal length f , and f_0 is the focal length of the central lens. The parameters D , f and f_0 must

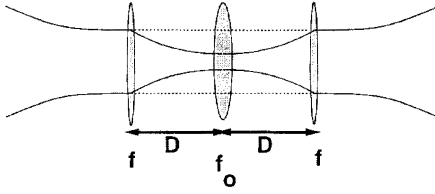


Fig. 1. Sketch of the beam inverter. The dotted line indicates the beam envelope in the yz -plane, which is not affected by the cylindrical lenses. The solid line gives the beam envelope in the xz -plane. Exact inversion occurs when the focal lengths are $f = D$, and $f_0 = D/4$.

be selected in such a way that the total difference in the Gouy phase for the x - and the y -direction amounts to π . Again, this phase difference must build up in the astigmatic region between the lenses. A sketch of the set-up and of a possible beam envelope is given in Fig. 1.

As one notices from Eqs. (15) and (16), the choice of the separation D and the focal length f in the case of the HG-LG converter depends on the specific properties of the incident beam, such as its Rayleigh range and the location of the focus. One might expect that this would also be true in the present case of an LG inverter. However, the following argument indicates that this is not so. Suppose that the LG inverter is constructed such that an incoming LG beam $\sim \exp(im\phi)$ with some specific Rayleigh range and focus position is properly converted to an output beam $\sim \exp(-im\phi)$. Since the lenses do not affect the y -dependence of the beam, the net result is a simple inversion of the x -dependence as compared with the output beam in the absence of the lenses. The relation between the LG output field u leaving the lens at $z = D$ and the unperturbed LG field u_0 at the same position in the absence of the lenses can therefore be expressed as $u(x, y, D) = u_0(-x, y, D)$ (20)

for any LG beam mode with the same Rayleigh range and focus position. Since these LG modes form a complete set, we can expand an arbitrary input beam $u(x, y, -D)$ arriving at the first lens as a linear combination of these LG modes. The conclusion is that the effect of the lens system can be expressed as in Eq. (20) for an arbitrary paraxial

input beam. This implies that the lens system is a universal beam inverter for paraxial beams.

In order to determine the parameters needed for LG-LG inversion, we express this conclusion in operator form. The net effect of this lens system on an arbitrary paraxial input beam is given by the ordered product of operators corresponding to the lenses and the intermittent free propagation, of the abbreviated form

$$\hat{C} = \hat{S}\hat{U}(D)\hat{S}_0\hat{U}(D)\hat{S}, \quad (21)$$

where $\hat{U}$ is the free-propagation operator (6), and $\hat{S}$ and $\hat{S}_0$ are operators for cylindrical lenses, with focal lengths f and f_0 , respectively. Since the lenses are focussing in the x -direction, one should use Eq. (9) with $\alpha = 0$. We introduce the x -inversion operator $\hat{M}_x$ by the identity

$$\hat{M}_x u(x, y) = u(-x, y). \quad (22)$$

Since $|u_0(D)\rangle$ is related to $|u(-D)\rangle$ by the free-propagation operator $\hat{U}(2D)$, the general condition (20) for the LG inverter is equivalent to the identity

$$|u(D)\rangle = \hat{C}|u(-D)\rangle = \hat{M}_x\hat{U}(2D)|u(-D)\rangle \quad (23)$$

for an arbitrary input beam $|u(-D)\rangle$. Hence, we must select the parameters D , f and f_0 of the lens system so that the operator identity

$$\hat{C} = \hat{M}_x\hat{U}(2D) \quad (24)$$

holds. The validity conditions for this identity can be obtained from the requirement that both sides lead to the same transformation of any operator $\hat{A}$, so that

$$\hat{C}^\dagger\hat{A}\hat{C} = \hat{U}^\dagger(2D)\hat{M}_x^\dagger\hat{A}\hat{M}_x\hat{U}(2D). \quad (25)$$

It is sufficient to substitute for the operator $\hat{A}$ the complete set of operators $\hat{X}$, $\hat{P}_x$, $\hat{Y}$ and $\hat{P}_y$. Since $\hat{M}_x$, $\hat{S}$ and $\hat{S}_0$ do not operate on the y -dependence of a mode function, the identity (25) trivially holds for $\hat{A} = \hat{Y}$ and $\hat{A} = \hat{P}_y$. For the other two cases $\hat{A} = \hat{X}$ and $\hat{A} = \hat{P}_x$, we use the transformation rules [14]

$$\hat{U}^\dagger(Z)\hat{X}\hat{U}(Z) = \hat{X} + \frac{Z}{k}\hat{P}_x, \quad (26a)$$

$$\hat{S}^\dagger\hat{P}_x\hat{S} = \hat{P}_x - \frac{k}{f}\hat{X}. \quad (26b)$$

When we take $\hat{A} = \hat{X}$ in Eq. (25), repeated application of Eq. (26) leads to the condition

$$\left(1 - \frac{D}{f_0} - \frac{2D}{f} + \frac{D^2}{ff_0}\right) \hat{X} + \frac{D}{k} \left(2 - \frac{D}{f_0}\right) \hat{P}_x = -\hat{X} - \frac{2D}{k} \hat{P}_x. \quad (27)$$

The resulting condition for the parameters of the lens system is given by

$$f_0 = D/4, \quad f = D. \quad (28)$$

In the same way one may verify that these relations (28) are also sufficient to ensure the validity of Eq. (25) for $\hat{A} = \hat{P}_x$. This serves as a check on the consistency of the arguments given above for the universality of the LG inverter.

The conclusion is that a symmetric system of three lenses characterized by the relations (28) has the effect of a simple inversion of the x -dependence of any incident paraxial beam. For Gaussian beams, this system serves as an exact π converter, regardless the values of the Rayleigh range and the location of the focus of the input beam. In the special case of an incident LG beam, it inverts the azimuthal dependence $\exp(im\phi) \rightarrow \exp(-im\phi)$, without affecting the radial dependence.

4. Frequency shift by a rotating LG inverter

We consider the case of an LG inverter that rotates about the z -axis at an angular velocity Ω . Strictly speaking, the axes of the three lenses should not be exactly parallel, in order to account for the retardation due to the finite travel time of the light. As one may conclude from Eq. (14), the orientation of the first lens at a given instant t must be parallel to the axis of the central lens a time D/c later, so that the output of the lens system has been affected by the three lenses with parallel orientations. For realistic rotation frequencies, this retardation effect can be safely ignored.

The input-output relation of the rotating lens system is expressed by

$$|u(D, t)\rangle = \hat{R}(\Omega t) \hat{C} \hat{R}^\dagger(\Omega t) |u(-D)\rangle. \quad (29)$$

When the parameters of the lens system obey Eqs. (28), the identity (24) is valid, and the effect of the rotating lenses is that the output beam is replaced by its mirror image with respect to a line that rotates in the xy plane at an angular velocity Ω . The effect of the rotation operator $\hat{R}(\Omega t)$ is that the azimuthal angle ϕ is replaced by $\phi - \Omega t$, whereas according to Eq. (24) the operator $\hat{C}$ leads to the simple inversion $\phi \rightarrow -\phi$, in addition to the free propagation over $2D$. By applying Eq. (29), we find that the only effect of the lenses is that in the output beam the azimuthal angle ϕ is substituted by $-\phi + 2\Omega t$.

We consider the case that the input beam is an LG beam with an azimuthal dependence $\sim \exp(im\phi)$. One notices from this substitution that the effect of the rotating inverter is twofold. First, the output beam has the azimuthal dependence $\sim \exp(-im\phi)$, as in the case of a stationary inverter. Second, the frequency of the output beam has the shifted value

$$\omega' = \omega - 2m\Omega. \quad (30)$$

In this special case of an LG input beam, the output is monochromatic. The field in the astigmatic region between the lenses, however, is strongly non-monochromatic, as illustrated by Eq. (13).

The frequency shift can be compared to the frequency shift -2Ω that occurs when a beam of circularly polarized light is converted by a rotating half-wave plate into the opposite circular polarization [2]. Frequency shifts arising from rotating birefringent plates have been interpreted as dynamical manifestations of the Pancharatnam phase [1]. The most straightforward interpretation results when one views the light beam in a co-rotating reference frame. In such a frame, the electric-field vector of circularly polarized light rotates at a frequency that is shifted by the angular velocity. This is just a rotational analogue of the (translational) Doppler effect, which also plays a role in the interpretation of the mechanical Faraday effect [15]. The double frequency shift occurs when the circular polarization is inverted by a rotating plate, in the same way as reflection from a moving mirror produces a double Doppler shift.

In the present case, it is the helicity of the wave fronts that is inverted rather than the light polarization. An LG beam with azimuthal quantum number m appears at a frequency that is shifted by $-m\Omega$ when it is viewed from a rotating frame, simply

because a single full turn around the axis corresponds to m oscillations. This frequency down-shift is doubled when the inverted LG beam is transformed back to the non-rotating frame. Hence the frequency shift $-2m\Omega$ allows a simple geometrical interpretation as a rotational Doppler shift.

The frequency shift of the beam by the rotating LG inverter corresponds to an exchange of energy from the light beam to the lens system. Suppose that the beam intensity in number of photons per unit time is N . Photons in the input beam have an (orbital) angular momentum $\hbar m$ per photon, whereas the photons in the output beam have angular momentum $-\hbar m$. This demonstrates that the system of lenses gains an angular momentum per unit time that is equal to

$$T = 2N\hbar m, \quad (31)$$

which therefore equals the combined torque on the lenses. Since the lens system is rotating at an angular velocity Ω , the work that the light torque exerts on the lenses corresponds to a gain of rotational energy per unit time

$$W = T\Omega. \quad (32)$$

The resulting power loss per photon $2\hbar m\Omega$ corresponds precisely to the frequency shift $\omega - \omega'$, that was obtained in Eq. (30). Of course, a similar dynamical interpretation can also be given for the frequency shift accompanying a rotating half-wave plate that inverts circular polarization, or, for that matter, for the Doppler shift of light reflected from a receding mirror.

5. Conclusion

With a system consisting of three parallel astigmatic lenses, one can construct an exact beam inverter, so that the output beam is the full mirror image of what the output beam would have been in the absence of the lenses. The system consists of two identical lenses in a confocal configuration, and a third lens located at their common focus. The focal length of this central lens is one fourth of the focal length of the two outer lenses. The system is insensitive to the specific properties of the input beam. The

only limitation is the validity of the paraxial approximation. It had been noted before that an inverter for LG beams can be realized only approximately with two lenses [11,13]. When the system is used to invert a Laguerre–Gaussian mode with azimuthal mode index m , the orbital angular momentum that is transferred to the lens system amounts to $2\hbar m$ per photon. This transfer of angular momentum shows up when the lens system is set into rotation with angular velocity Ω , in that the output frequency is downshifted by the amount $2m\Omega$. The corresponding power loss of the light beam corresponds to the mechanical work exerted by the torque on the lenses. Since both the input and the output beam have a circular nature, there can be no net torque on the two outer lenses, which implies that the full angular momentum transfer is picked up by the central lens. This effective rotational Doppler effect allows both a geometrical and a dynamical interpretation. When the input beam is not an exact Laguerre–Gaussian beam, the output beam acquires a series of sidebands. This means that the monochromaticity of the output may serve as a check on the azimuthal dependence of the input beam.

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Poincaré-sphere equivalent for light beams containing orbital angular momentum

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The polarization state of a light beam is related to its spin angular momentum and can be represented on the Poincaré sphere. We propose a sphere for light beams in analogous orbital angular momentum states. Using the Poincaré-sphere equivalent, we interpret the rotational frequency shift for light beams with orbital angular momentum [Phys. Rev. Lett. **80**, 3217 (1998)] as a dynamically evolving geometric phase. © 1999 Optical Society of America
OCIS code: 350.1370.

Light beams possess a spin angular momentum that is associated with their polarization state. It is less well known that light beams also possess an orbital angular momentum that is associated with their azimuthal phase structure. This orbital angular momentum is independent of the polarization state of the light and therefore independent of the spin. Both spin and orbital angular momentum have been experimentally transferred to matter, causing rotation.¹⁻³

Right- and left-handed circularly polarized light, the spin eigenstates, have a spin angular momentum of $+\hbar$ and $-\hbar$ per photon, respectively. In 1992, Allen *et al.* found that Laguerre–Gaussian modes, which have an azimuthal phase term of $\exp(i l \phi)$, are orbital angular momentum eigenstates and carry an orbital angular momentum of $l\hbar$ per photon.⁴ Laguerre–Gaussian modes are denoted LG_p^l , where l and p are the azimuthal and the radial mode indices, respectively. The quantity $N = 2p + |l|$ is referred to as the order of the mode.⁵

The polarization state of a monochromatic light beam can be completely characterized in terms of the Stokes parameters⁶

$$\begin{aligned} p_1 &= \frac{I_{0^\circ} - I_{90^\circ}}{I_{0^\circ} + I_{90^\circ}}, \\ p_2 &= \frac{I_{45^\circ} - I_{135^\circ}}{I_{45^\circ} + I_{135^\circ}}, \\ p_3 &= \frac{I_{\text{right}} - I_{\text{left}}}{I_{\text{right}} + I_{\text{left}}}, \end{aligned} \quad (1)$$

where I_{0° , I_{45° , I_{90° , and I_{135° are the intensities of the light recorded through various orientations of linear polarizers and I_{right} and I_{left} are the intensities of the circularly polarized components in the beam. For a completely polarized light beam the squares of the Stokes parameters add up to unity, i.e.,

$$p_1^2 + p_2^2 + p_3^2 = 1. \quad (2)$$

Consequently, the Stokes parameters are the Cartesian coordinates of a space in which any completely polarized light beam is represented by a point on

a sphere with unit radius around the origin. This sphere is known as the Poincaré sphere [Fig. 1(a)] and has proved to be a useful tool in dealing with transformations of the polarization state.⁶

The north and south poles of the Poincaré sphere represent the spin eigenstates, left- and right-handed circularly polarized light, respectively. Any state of complete polarization, and therefore any point on the Poincaré sphere, can be described as a superposition of left- and right-handed circular polarizations. For example, linearly polarized light is a superposition of equal intensities of left- and right-handed circularly polarized light; the relative phase of the superposition determines the orientation of the linear polarization.

We construct an analogous sphere for superpositions of left- and right-handed Laguerre–Gaussian modes with azimuthal phase terms of $\exp(\pm i l \phi)$, respectively, which are orbital angular momentum eigenstates. These Laguerre–Gaussian modes (Fig. 2) are denoted LG_0^{+1} and LG_0^{-1} and possess an orbital angular momentum of $+\hbar$ and $-\hbar$ per photon, respectively. Their superpositions form structurally stable beams (modes) of mode order⁷ $N = 1$. Therefore, as every point on the Poincaré-sphere analog corresponds to a mode of order 1, and also every mode of order 1 corresponds to a point on the Poincaré-sphere analog, we refer to this sphere as the sphere of first-order modes. For example, a superposition of equal intensities of left- and right-handed Laguerre–Gaussian modes forms a Hermite–Gaussian mode with indices $m = 1$ and $n = 0$, denoted $HG_{1,0}$, which is of order⁵ $N = m + n = 1$. The

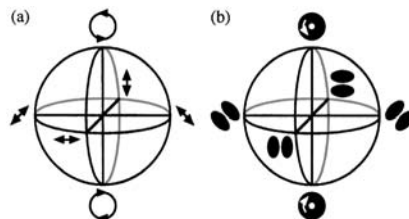


Fig. 1. (a) Poincaré sphere and (b) sphere of first-order modes. Some commonly encountered polarization states and modes are indicated.

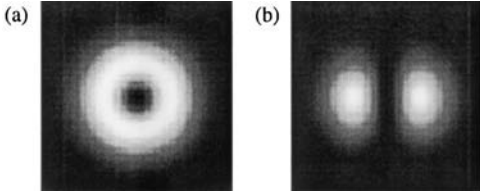


Fig. 2. Intensity cross section of (a) $LG_0^{\pm 1}$ modes and (b) a $HG_{1,0}$ mode. The modes LG_0^{-1} and LG_0^{+1} differ only in the handedness of their azimuthal mode structure.

relative phase of the superposition determines the orientation of the mode.

In analogy to the Stokes parameters, we define a set of parameters, o_1, o_2, o_3 , for the newly proposed sphere:

$$\begin{aligned} o_1 &= \frac{I_{HG_{1,0}^{0^\circ}} - I_{HG_{1,0}^{90^\circ}}}{I_{HG_{1,0}^{0^\circ}} + I_{HG_{1,0}^{90^\circ}}}, \\ o_2 &= \frac{I_{HG_{1,0}^{45^\circ}} - I_{HG_{1,0}^{135^\circ}}}{I_{HG_{1,0}^{45^\circ}} + I_{HG_{1,0}^{135^\circ}}}, \\ o_3 &= \frac{I_{LG_0^1} - I_{LG_0^{-1}}}{I_{LG_0^1} + I_{LG_0^{-1}}}, \end{aligned} \quad (3)$$

where $I_{HG_{1,0}^\alpha}$ stands for the intensity of the Hermite–Gaussian $m = 1, n = 0$ mode at angle α that is present in the beam and $I_{LG_0^{\pm 1}}$ stands for the intensities of the Laguerre–Gaussian $l = \pm 1, p = 0$ modes in the beam. Writing all the intensities as the square of the moduli of the corresponding mode functions, and expanding all the mode functions in terms of Laguerre–Gaussian $l = \pm 1, p = 0$ mode functions,⁵ we can show that

$$o_1^2 + o_2^2 + o_3^2 = 1. \quad (4)$$

Starting with left-handed circularly polarized light, a quarter-wave plate transforms the light to a linear polarization at 45° to the axis of the wave plate. On the Poincaré sphere, this transformation is represented by a move from the north pole to a point on the equator, the longitude of which depends on the orientation of the linear polarization.⁸ Rotation of the wave plate through an angle α advances the longitude of the transformation by an angle 2α . Similarly, a half-wave plate transforms from left-handed to right-handed circular polarization. This transformation is presented by a move from pole to pole along a great circle, the longitude of which depends on the rotation angle of the wave plate.⁸ This dependence on the rotation angle can be demonstrated if we consider the half-wave plate as two identical quarter-wave plates: As before, the state after the first wave plate is represented on the equator, and its longitude depends on the rotational alignment of the fast axis of the wave plate. Rotation of the wave plate through an angle α advances the longitude of the trajectory taken during the transformation by 2α .

In a fashion similar to that for a birefringent wave plate, which controls the relative phase between

two linear polarizations at 90° to each other, the relative phase of two Hermite–Gaussian $m = 1, n = 0$ modes at 90° to each other (Fig. 3) can be controlled by an arrangement of cylindrical lenses. Beijersbergen *et al.*⁵ detailed the design and operation of cylindrical-lens mode converters that utilize changes in relative Gouy phase to transform Hermite–Gaussian modes into Laguerre–Gaussian modes and vice versa. Both π converters and $\pi/2$ converters exist that perform tasks analogous to those performed by half-wave and quarter-wave plates, respectively. One may best appreciate the analogy by considering both the wave plates and the cylindrical lenses as means of introducing phase shifts between two orthogonal components of the beam.

In 1979 Garetz and Arnold⁹ demonstrated that when circularly polarized light is transmitted through a rotating half-wave plate a frequency shift results. The shift is equal to twice the rotation frequency of the wave plate. Although the shift can be simply explained by Jones matrix polarization analysis, a similar experiment was also performed and explained in terms of a dynamic geometric (or Berry) phase by Simon *et al.*⁸ In their experiment a combination of wave plates was used to transform the polarization state of the light around a closed loop on the Poincaré sphere. The resulting geometric phase shift was equal to half the solid angle enclosed by the loop and, in addition, to the constant phase shift that was due to the optical thickness of the components. Uniform rotation of one of the wave plates gave a constant rate of change of the enclosed solid angle, resulting in a frequency shift of the light beam.

Recently a similar experiment was performed with light beams containing orbital angular momentum.¹⁰ When it is transmitted through a rotating π converter, the frequency of a $LG_0^{\pm 1}$ mode is shifted by twice the rotation frequency of the π converter. The mode transformation in the π converter can be interpreted as a geometric phase effect⁷; we interpret the frequency shift that occurs with transmission of the mode through a rotating π converter as evolution of this geometric phase in time.

Figure 4 illustrates the equivalence of the geometric phase shift for light beams containing either spin or orbital angular momentum after transmission through two half-wave plates or two π converters, respectively. In both cases the geometric phase shift is equal to half

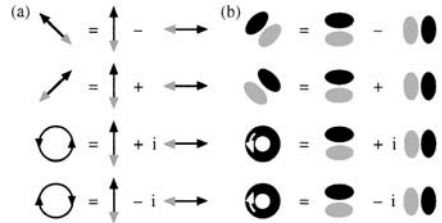


Fig. 3. Decompositions of (a) various polarization states and (b) first-order modes in terms of orthogonal linear polarization states and orthogonal Hermite–Gaussian $m = 1, n = 0$ modes, respectively. The black and gray shading indicates the relative phase of the components.

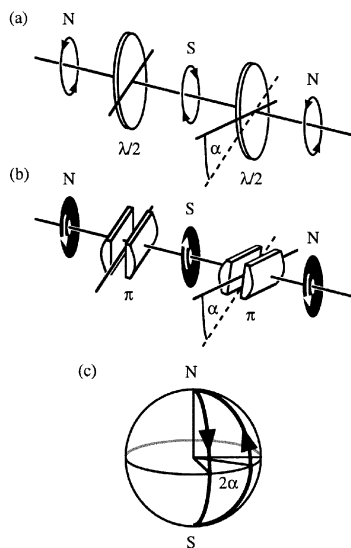


Fig. 4. Generation of evolving geometric phases for light beams containing (a) spin and (b) orbital angular momentum. (c) Path taken on the Poincaré sphere or the sphere of first-order modes.

the solid angle enclosed by the loop on the surfaces of the respective spheres. Therefore, for a change in the angular orientation between the two optical components of α , the resulting geometric phase shift is 2α . Thus, in both cases, the frequency of the light beam is shifted by twice the rotation frequency of the optical component.

We have proposed a Poincaré-sphere equivalent for light beams containing orbital angular momentum, called the sphere of first-order modes. On this sphere cylindrical-lens mode converters perform transformations analogous to those performed by wave plates on the Poincaré sphere. For the geometric phase accompanying transformations of first-order modes, the sphere of first-order modes is what the Poincaré sphere is for the geometric phase accompanying transformations of the polarization state.

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Rotational Frequency Shift

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The notion of the rotational frequency shift, an analog of the Doppler shift, is introduced. This new frequency shift occurs for atomic systems that lack rotational invariance, but have stationary states in a rotating frame. The rotational frequency shift is given by the scalar product of the angular velocity and the angular momentum of the emitted photon in full analogy with the standard Doppler shift which is given by the scalar product of the linear velocity of the source and the linear momentum of the photon. The rotational frequency shift can be observed only in a Mössbauer-like regime when the angular recoil is negligible. [S0031-9007(97)02782-8]

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The purpose of this Letter is to describe a new effect: the frequency shift of emitted (or absorbed) photons that is due to the rotation of the radiating system. This rotational frequency shift (RFS) is a close analog of the standard, first-order Doppler shift (the latter might be called a translational frequency shift) but there are two important differences.

First, the dynamical laws are invariant under a uniform translation, but they are not invariant under a uniform rotation. Therefore, the differences between the energy levels are not changed by a uniform translation but they are dynamically modified by a uniform rotation, owing to the centrifugal and Coriolis forces. Hence, the frequency of a photon emitted (absorbed) by a system moving with constant velocity is modified *only* by the Doppler shift, while in the case of a rotating system the photon frequency is modified *both* by the changes in the energy levels and by the RFS. For rotationally invariant systems, these two effects completely cancel each other as a result of the conservation of angular momentum, and the observed photon frequency is unchanged, as one would have anticipated.

Second, under normal conditions the recoil corrections are much more significant for the RFS than for the Doppler shift. In order to observe the RFS one must work in a Mössbauer-like regime. The atomic system must be embedded in a larger structure that will provide the angular momentum of the emitted photon.

The RFS should not be confused with the ordinary linear Doppler shift observed for rotating objects (for example, stars or galaxies) that is due to the instantaneous linear motion of the emitter. This linear Doppler shift is maximal in the plane of rotation while the RFS is maximal along the angular velocity, that is in the direction perpendicular to the instantaneous velocity. Thus the RFS competes with the quadratic Doppler shift rather than with the linear Doppler shift.

Our discussion will be based on the nonrelativistic Schrödinger equation for the atomic system with a time

dependent Hamiltonian. The photon emission will be treated in the first order of perturbation theory. Our derivation of the RFS will be done in parallel with the calculation of the standard Doppler shift in order to exhibit their similarities and differences.

As a generic model of a radiating system we shall consider an electron bound by a potential $V(t)$ and interacting via minimal coupling with the quantized electromagnetic field. We assume that the time dependence of the potential is of the form $V(t) = V(\mathbf{r}(t))$. For a uniform translation and a uniform rotation, $\mathbf{r}(t)$ is given by

$$\text{Translation } x(t) = x, \quad y(t) = y, \quad z(t) = z - vt, \quad (1)$$

$$\begin{aligned} \text{Rotation } x(t) &= x \cos(\Omega t) + y \sin(\Omega t), \\ y(t) &= -x \sin(\Omega t) + y \cos(\Omega t), \\ z(t) &= z. \end{aligned} \quad (2)$$

Since the time dependence of the potential is prescribed, our analysis will be applicable only to those cases when the change in the uniform motion of the potential caused by the photon emission (absorption) can be disregarded. This means that we neglect all recoil corrections.

The time dependence of the state vector $\Psi(t)$ is determined by the evolution equation (in atomic units)

$$\begin{aligned} i\partial_t \Psi(t) &= H(t)\Psi(t) \\ &= \left[\frac{1}{2} \mathbf{p}^2 + V(t) + H_F + H_I \right] \Psi(t), \end{aligned} \quad (3)$$

where H_F ,

$$\begin{aligned} H_F &= \frac{1}{2} \int d^3x (\mathbf{D}^2/\epsilon_0 + \mathbf{B}^2/\mu_0) \\ &= \sum_{\lambda} \int d^3k \omega(\mathbf{k}) a_{\lambda}^{\dagger}(\mathbf{k}) a_{\lambda}(\mathbf{k}) \\ &= \sum_{JM\lambda} \int d\omega \omega a_{JM\lambda}^{\dagger}(\omega) a_{JM\lambda}(\omega), \end{aligned} \quad (4)$$

is the free field Hamiltonian and H_I ,

$$H_I = \mathbf{p} \cdot \mathbf{A}(\mathbf{r}) + \frac{1}{2} \mathbf{A}^2(\mathbf{r}), \quad (5)$$

is the interaction Hamiltonian.

In order to determine the spectrum of emitted photons, we construct the transition amplitude A_{fi} ,

$$A_{fi} = \langle \Psi_f | U(t, t_0) | \Psi_i \rangle. \quad (6)$$

The evolution operator $U(t, t_0)$ satisfies the equation

$$i \partial_t U(t, t_0) = H(t) U(t, t_0), \quad (7)$$

and the initial condition $U(t_0, t_0) = 1$. Next, we introduce a unitary transformation $\exp(iGt)$ to get rid of the time dependence of the Hamiltonian $H(t)$, transforming it back to time $t = 0$,

$$H(0) = e^{iGt} H(t) e^{-iGt}. \quad (8)$$

The operators G are the generators of uniform translations and uniform rotations,

$$\text{Translation } G = v P_z, \quad (9)$$

$$\text{Rotation } G = \Omega M_z, \quad (10)$$

where P_z and M_z denote the operators of the z component of the *total* linear momentum and angular momentum (including the electromagnetic field),

$$P_z = p_z + \sum_{\lambda} \int d^3k \, k_z a_{\lambda}^{\dagger}(\mathbf{k}) a_{\lambda}(\mathbf{k}), \quad (11)$$

$$M_z = x p_y - y p_x + \sum_{J M \lambda} \int d\omega \, M a_{J M \lambda}^{\dagger}(\omega) a_{J M \lambda}(\omega). \quad (12)$$

Note that the Hamiltonians H_F and H_I are invariant under translations and rotations and, therefore, they are not changed by the unitary transformations (8). One may check that the solution of the evolution equation (7) satisfying the initial condition has the form

$$U(t, t_0) = e^{-iGt} e^{-i(H_0 + H_I)(t-t_0)} e^{iGt_0}, \quad (13)$$

where

$$H_0 = \frac{1}{2} \mathbf{p}^2 + V(0) + H_F - G. \quad (14)$$

In the first order of perturbation theory, we obtain

$$\begin{aligned} U(t, t_0) &\approx e^{-iGt} e^{-iH_0(t-t_0)} e^{iGt_0} \\ &- i e^{-iGt} \int_{t_0}^t d\tau e^{-iH_0(t-\tau)} \\ &\quad \times \mathbf{p} \cdot \mathbf{A}(\mathbf{r}) e^{iH_0(t_0-\tau)} e^{iGt_0}. \end{aligned} \quad (15)$$

Upon inserting this expression into the formula (6) for the transition amplitude, we get

$$\begin{aligned} A_{fi} &\approx \langle \Phi_f | e^{-iH_0(t-t_0)} | \Phi_i \rangle \\ &- i \langle \Phi_f | \int_{t_0}^t d\tau e^{-iH_0(t-\tau)} H_I e^{iH_0(t_0-\tau)} | \Phi_i \rangle, \end{aligned} \quad (16)$$

where

$$|\Phi_i\rangle = e^{iGt_0} |\Psi_i\rangle, \quad |\Phi_f\rangle = e^{iGt} |\Psi_f\rangle. \quad (17)$$

The transition amplitude (16) will lead to the Fermi Golden Rule for the transition rate provided the auxiliary vectors $|\Phi_{i,f}\rangle$ are chosen as eigenvectors of H_0 . In order to study spontaneous emission, we shall assume that the initial state comprises the excited state of the electron and the vacuum state of the field and the final state comprises the ground state of the electron and a one photon state. The corresponding eigenvectors $|\Phi_{i,f}\rangle$ have a product form and they satisfy the following eigenvalue equations:

$$H_0 |\phi_e(\mathbf{r})\rangle |0\rangle = E_e |\phi_e(\mathbf{r})\rangle |0\rangle, \quad (18)$$

$$H_0 |\phi_g(\mathbf{r})\rangle |1_{ph}\rangle = (E_g + \omega - \Delta) |\phi_g(\mathbf{r})\rangle |1_{ph}\rangle, \quad (19)$$

where $\phi_e(\mathbf{r})$ and $\phi_g(\mathbf{r})$ are the eigenfunctions of the electronic part H_E of H_0 ,

$$\text{Translation } H_E = \frac{1}{2} \mathbf{p}^2 + V(0) - v p_z, \quad (20)$$

$$\text{Rotation } H_E = \frac{1}{2} \mathbf{p}^2 + V(0) - \Omega(x p_y - y p_x), \quad (21)$$

corresponding to the eigenvalues E_e and E_g . The shift Δ is an eigenvalue of the photonic part of the generator G [cf. Eqs. (11) and (12)] and is equal to $v k_z$ in the case of translation or to ΩM in the case of rotation. The vectors $|\Phi_{i,f}\rangle$ represent stationary states; the electronic probability densities $|\phi_{e,g}(\mathbf{r})|^2$ are time independent. The original vectors $|\Psi_{i,f}\rangle$, related to the eigenvectors $|\Phi_{i,f}\rangle$ by the transformation $\exp(-iGt)$ represent nonstationary states. Since the transformation $\exp(-iGt)$ acting on the electronic wave function replaces its argument $\mathbf{r}$ by $\mathbf{r}(t)$, the electronic wave functions depend on time not only through a phase factor $\exp(-iEt)$ but also through $\mathbf{r}(t)$. These states may be called quasistationary since the corresponding electronic probability densities $|\psi_{e,g}(\mathbf{r}(t))|^2$ undergo a uniform motion according to the formulas (1) and (2).

The spectrum of the emitted photons can be determined from the standard formula for the decay rate w_{if} applied to the amplitude (16)

$$w_{if} = 2\pi |\langle \Phi_f | H_I | \Phi_i \rangle|^2 \delta(E_e - E_g - \omega + \Delta). \quad (22)$$

Hence the frequency of the emitted photon is shifted by Δ . For a uniform translation, Δ represents the standard Doppler shift and for a uniform rotation, Δ represents the *rotational frequency shift*,

$$\text{Translation (Doppler)} \quad \omega = E_e - E_g + v k_z, \quad (23)$$

$$\text{Rotation (RFS)} \quad \omega = E_e - E_g + \Omega M. \quad (24)$$

In the case of a uniform translation the electronic energy difference $E_e - E_g$ is exactly the same as for a system at rest because the electronic part of the Hamiltonian H_0 is unitarily equivalent (up to a constant) to the corresponding Hamiltonian at rest

$$\begin{aligned} e^{ivz} \left[\frac{1}{2} \mathbf{p}^2 + V(0) - v p_z \right] e^{-ivz} \\ = \frac{1}{2} \mathbf{p}^2 + V(0) - \frac{1}{2} v^2. \end{aligned} \quad (25)$$

Therefore for a uniform translation the only change in the spectrum of the emitted photons is the Doppler shift. This is simply the result of the Galilean invariance of nonrelativistic quantum mechanics. The situation is quite different for a uniform rotation. The spectrum of the photons emitted by a rotating source is not only shifted by $\Delta = \Omega M$ but it is also, in general, modified due to changes of the energy differences $E_e - E_g$.

This modification is especially simple for rotationally invariant potentials when M_z commutes with the Hamiltonian H_0 . In this case the wave functions $\phi_e(\mathbf{r})$ and $\phi_g(\mathbf{r})$ can be chosen as eigenfunctions of the z component of the electronic angular momentum belonging to certain eigenvalues m_e and m_g . Then the change in the energy difference caused by the rotation is equal to $\Omega(m_e - m_g)$. This change cancels *completely* the RFS owing to the conservation of angular momentum and no shift is observed in the laboratory frame. Therefore, the RFS cannot be seen in isolated atoms but it might occur, in principle, in isolated molecules or in atoms placed in an environment that destroys the rotational symmetry.

Even when the potential is not rotationally invariant, the consequences of the RFS are not always significant. In particular, when the rotating body is of macroscopic dimensions the RFS is much smaller, in general, than other effects. For example, in the experiment [1] designed to measure the transverse Doppler shift the emitter and the absorber were mounted on a rotating disk. Even though the angular velocity was large enough to observe the quadratic Doppler shift, the RFS calculated for the conditions of this experiment could not have been seen (it is only about 10^{-4} of the quadratic Doppler shift). The RFS is so small in this case because the ratio of the wavelength to the linear dimensions of the rotating system is exceedingly small. The RFS might be seen for macroscopic rotating systems when the emitted radiation is in the radio-wave range, as is the case for transitions involving nuclear magnetic moments. Taking the frequency of such a transition in the range of 10^8 Hz and assuming an ultrafast centrifuge rotating at 10^3 Hz, one obtains the RFS shift of the order of 10^{-5} of the resonance frequency. Such shifts might be detectable. In order to observe the RFS in the optical domain, the rotational frequency must be much higher and that can be achieved only when the rotating systems are of atomic dimensions.

The simplest model of a rotating atomic system with broken rotational symmetry is an electron bound by a rotating asymmetric harmonic potential in two dimensions,

$$V(t) = \frac{1}{2} [\mu x^2(t) + \nu y^2(t)], \quad (26)$$

where μ and ν are two arbitrary real parameters and the dependence of x and y on time is given by the formulas (2). An especially interesting case is when one of the two oscillators is inverted (one of the parameters μ or ν is negative). Such a model has been recently used to predict the existence and to describe the main features of Trojan wave packets of Rydberg electrons in atoms [2] and in

rotating molecules with large electric dipole moments [3]. Since all more elaborate numerical calculations [4–7] fully confirmed the validity of the estimates based on the harmonic approximation, we shall use the model with the potential (26) to study the significance of the RFS in realistic situations.

The Trojan states of Rydberg electrons in hydrogenic atoms are described by nonspreading localized wave packets circling on a large orbit around the nucleus. Their stability is due to an interplay between the centrifugal force and the Coriolis force acting in the rotating frame and the electric field of a circularly polarized electromagnetic wave propagating in the direction perpendicular to the orbit. They were named by us the Trojan states because their mathematical description is similar to the description of the orbits of Trojan asteroids in the Sun-Jupiter system [2]. The Trojan states of electrons offer a perfect example of an atomic system that will exhibit the RFS since in the laboratory frame the interaction with the electric field of the wave is described (in the dipole approximation) by the rotating potential $V(t) = \mathcal{E}[x \cos(\Omega t) + y \sin(\Omega t)]$. The description of the Trojan states in the harmonic approximation [2] leads to the Hamiltonian in which the parameters μ and ν are determined by the frequency of the wave Ω and by the parameter q that measures the ratio between the Coulomb force and the centrifugal force, $\mu = -2q\Omega^2$, $\nu = q\Omega^2$. The frequencies $\omega_{\pm}$ in this case are

$$\omega_{\pm} = \Omega \sqrt{(2 - q \pm \sqrt{9q^2 - 8q})/2}. \quad (27)$$

They are real for $8/9 \leq q \leq 1$.

We shall use now the harmonic oscillator model to exhibit the main features of the RFS for realistic atomic systems. The electronic part H_E of the time independent Hamiltonian for the potential (26) is in this case

$$H_E = \frac{1}{2} \mathbf{p}^2 + \frac{1}{2} (\mu x^2 + \nu y^2) - \Omega (xp_y - yp_x). \quad (28)$$

This Hamiltonian describes two harmonic oscillators with the following characteristic frequencies

$$\omega_{\pm} = \sqrt{(\mu + \nu + 2\Omega^2 \pm \sqrt{(\mu - \nu)^2 + 8(\mu + \nu)\Omega^2})/2} \quad (29)$$

that determine the spacing between the energy levels. These frequencies are real when the parameters μ , ν , and Ω satisfy the conditions

$$\mu \geq \Omega^2 \quad \text{and} \quad \nu \geq \Omega^2, \quad (30)$$

or

$$\begin{aligned} -3\Omega^2 \leq \mu \leq \Omega^2, \quad -3\Omega^2 \leq \nu \leq \Omega^2, \\ \text{and } (\mu - \nu)^2 + 8(\mu + \nu)\Omega^2 \geq 0. \end{aligned} \quad (31)$$

In a degenerate case of an isotropic potential, when $\mu = \nu$, the frequencies $\omega_{\pm}$ are linear functions of the angular velocity $\omega_{\pm} = \sqrt{\mu} \pm \Omega$. This is a special example of a rotationally invariant potential, when the RFS is exactly

canceled by the dynamical modification of the electronic energy difference. For anisotropic potentials the energy differences are nonlinear functions of Ω and a complete cancellation between the RFS and the dynamical modifications of the energy differences is impossible.

The RFS that occurs in the emission of photons during transitions between the atomic Trojan states has dramatic consequences. The frequencies (24) of the photons emitted during the transitions between the neighboring energy levels of the oscillators, as seen by the observer in the laboratory frame, are

$$\omega = \omega_{\pm} + \Omega M. \quad (32)$$

The observed photon frequency is drastically shifted by the RFS; the shift *exceeds* the original transition frequency in the rotating frame. This huge effect is due to the fact that the rotation is not just a small modification of the dynamics, but the mere existence of the Trojan states depends on it. Since both frequencies $\omega_{\pm}$ are smaller than Ω , the RFS has a very large effect in this case. The emission of photons with $M = -1$ is forbidden. This means that the photons emitted spontaneously by the Trojan electrons have the same projection of the angular momentum on the z axis as the photons constituting the strong wave that is driving the system. At this point we would like to stress that the RFS and the transfer of the angular momentum to the emitted photon must not in any direct way be attributed to the photons that make the circularly polarized wave. The same frequency shift is obtained for all rotating potentials, regardless of their nature. For example, it would appear for nonspherical rotating nuclei, provided one can treat the rotating nuclear potential as fully prescribed.

In order to illustrate these general results with specific numbers, we shall consider a hydrogen atom driven by the circularly polarized wave with the frequency of 200 GHz and the field amplitude of 1930 V/m, the same as in Ref. [2]. In this case, the separations between the neighboring oscillator levels in the rotating frame are 1.92×10^{11} Hz and 0.66×10^{11} Hz. The frequencies of the circularly polarized radiation observed in the laboratory frame will be shifted by 2×10^{11} Hz and they will be equal to 3.92×10^{11} Hz and 2.66×10^{11} Hz, respectively. Both frequencies are quite different from the frequency of the driving wave.

Our derivation of the RFS depends crucially on the validity of the no-recoil approximation. This assumption is satisfied very well for the Doppler shift but the effect of recoil in the rotational motion is much more significant than in the translational motion. Usually, one may disregard the perturbation of the translational motion by the emission of the photon because the relative change in the velocity is of the order of 10^{-4} . In contrast, the relative change of the angular velocity of a molecule caused by an emission of a photon, as argued below, is several orders of magnitude

larger. The principle of energy equipartition gives the relation between the angular velocity and the linear velocity $\Omega = v/R$, where R is the effective radius of the molecule. Therefore the relative changes of the angular velocity and the linear velocity are related by a large factor, the ratio of the wave length to the molecular radius,

$$\delta\Omega/\Omega \approx (1/kR)(\delta v/v), \quad (33)$$

where k is the photon wave vector. This means that for an isolated molecule the angular recoil should be taken into account. The whole molecule, not just the electron undergoing the radiative transition, must be considered as a radiating system. For the molecule, treated as a closed system, the RFS is unobservable because the total angular momentum is conserved and we end up in the category of rotationally invariant systems. However, for strongly driven atoms or molecules, as in the case of atomic Trojan states, a Mössbauer-like regime is reached since the external field sustains the uniform rotational motion. Under such conditions, the angular recoil can be disregarded since the angular momentum of the emitted photon is provided by the macroscopic field.

The standard explanation of the Doppler shift is based on the kinematics of a special theory of relativity [8,9]. The RFS may also be explained in a similar vein but one must apply general theory of relativity because the rotating coordinate frame is not inertial. One may use in this case the known transformation properties of the Maxwell equations under general coordinate transformations to relate the frequency of the photon emitted in the rotating frame to the frequency observed in the laboratory frame. However, the RFS is always accompanied by the modifications of the emitter dynamics due to rotation and the net frequency shift can be observed *only for quantum mechanical systems* when the electronic Hamiltonian does not commute with the generator of rotations.

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Azimuthal Doppler shift in light beams with orbital angular momentum

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Abstract

We show that an atom moving in a light beam with orbital angular momentum experiences an azimuthal shift in the resonant frequency in addition to the usual axial Doppler and recoil shifts. For a Laguerre-Gaussian beam characterised by an orbital angular momentum quantum number l , the shift is lV_ϕ/r where r is the radial atomic position and V_ϕ the azimuthal component of velocity. The predicted shift could play a significant role in interactions between atoms and standing light fields in cooling experiments as well as in ion traps.

There has been a good deal of interest in the mechanical effects that light can exert on material systems. Most attention in the last decade or so [1] has been addressed to the *translational* effects of light.

The best known experiment on the *rotational* aspects of light is that of Beth [2], who measured the torque experienced by a birefringent plate subject to a beam of circularly polarised light. In Beth's experiment the relevant effect is attributable to the spin angular momentum. Recent emphasis on rotational effects has stemmed from work by Allen et al. [3] who suggested that paraxial beams, in particular Laguerre-Gaussian beams, possessing orbital angular momentum should influence the angular motion of a rigid body. They proposed an experiment that exploits the transfer of angular momentum in the process of converting a paraxial beam of angular momentum quantum number l into a Hermite-Gaussian beam which has no orbital angular momentum. A subsequent report by Beijersbergen et al. [4] discussed the actual design, and demonstrated the operation, of a converter

which could experience a resultant torque.

There has been speculation recently about the possible rotational effects that atoms may experience when immersed in light beams possessing orbital angular momentum [3,5]. A moving atom interacting with light experiences changes in its gross motion due to an associated radiation force. Such effects have in the past only been considered in the context of plane waves. The way in which the intrinsic orbital angular momentum of the beam manifests itself in the behaviour of the atoms is the main concern of this article. We focus on a motion-induced effect that lends itself to direct experimental study, namely the shift in the atomic resonance. The aim is to quantify the changes that occur in the Doppler effect for atoms interacting with light beams with orbital angular momentum.

The salient features of the theory may be discussed with respect to a two-level atom, of resonant frequency ω_0 , interacting with a light mode of frequency ω . In the electric dipole and rotating wave approximations the Hamiltonian may be written [6]

$$H = \hbar\omega_0\pi^\dagger\pi + \mathbf{P}^2/2M + U(\mathbf{R}) + \hbar\omega a^\dagger a - i\hbar[\pi^\dagger a f(\mathbf{R}) - f^*(\mathbf{R}) a^\dagger \pi], \quad (1)$$

where $\mathbf{P}$ and $\mathbf{R}$ are the momentum and position vectors of the centre of mass with total mass M , π and $\pi^\dagger$ are ladder operators characterising the internal two-level system, and a and $a^\dagger$ are, respectively, the annihilation and creation operators of the light field. The inclusion of the potential $U(\mathbf{R})$ allows the general consideration of the atom, whether free or confined. The operator $f(\mathbf{R})$ stems from the electric dipole interaction $-\mathbf{d} \cdot \mathbf{E}(\mathbf{R})$ and we may write

$$f(\mathbf{R}) = N(\mathbf{D}_{12} \cdot \mathbf{E}(\mathbf{R})), \quad (2)$$

where $\mathbf{E}$ is the electric field vector associated with the mode, N is a normalisation factor and $\mathbf{D}_{12}$ is the dipole matrix element.

The above Hamiltonian (Eq. (1)) is the basis for a direct quantum mechanical derivation of the Doppler shift [7]. The time evolution of the operator $f(\mathbf{R}(t))$ stems from its Heisenberg equation of motion which, together with Eq. (1), gives

$$\begin{aligned} \frac{df}{dt} &= \frac{i}{\hbar} [H, f(\mathbf{R}(t))] \\ &= \frac{1}{2M} (\mathbf{P} \cdot \nabla f(\mathbf{R}(t)) + \nabla f(\mathbf{R}(t)) \cdot \mathbf{P}). \end{aligned} \quad (3)$$

As $U(\mathbf{R})$ commutes with $f(\mathbf{R})$, Eq. (3) holds irrespective of the explicit form of the potential $U(\mathbf{R})$.

The solution of Eq. (3) can only be obtained if the time dependence of both $\mathbf{R}$ and $\mathbf{P}$ have been specified. However we need only evaluate f to leading order in the interaction, which amounts to depending on the initial values of $\mathbf{R}$ and $\mathbf{P}$ of the atomic operators. We therefore obtain the leading solution of Eq. (3) in the form

$$\begin{aligned} f(\mathbf{R}, t) &\approx f_0 \exp \left[\left(\frac{(\mathbf{P} \cdot \nabla f + \nabla f \cdot \mathbf{P})}{2Mf} \right)_0 t \right] \\ &= f_0 \exp(i\delta t), \end{aligned} \quad (4)$$

where the subscript zero denotes operators at the initial time $t = 0$. The change in energy due to the atomic gross motion is identified in terms of the corresponding frequency operator δ appearing in Eq. (4)

$$\delta = -\frac{i}{2M} \left(\frac{\mathbf{P} \cdot \nabla f + \nabla f \cdot \mathbf{P}}{f} \right)_0. \quad (5)$$

In what follows it is convenient to drop the zero subscript from the operators $\mathbf{R}$ and $\mathbf{P}$, but retain it in f_0 .

By restricting attention to the case of a linearly polarised plane wave the frequency shift δ may be readily shown to contain the familiar form of the Doppler shift as well as the recoil shift. For a plane wave of wavevector $\mathbf{k}$ and polarisation $\boldsymbol{\epsilon}$ we have $f_0 = N(\mathbf{D}_{12} \cdot \boldsymbol{\epsilon}) \exp(-i\mathbf{k} \cdot \mathbf{R})$ and we then obtain from Eq. (5) the well known result

$$\delta = -\mathbf{k} \cdot \mathbf{V} + \frac{\hbar k^2}{2M}, \quad (6)$$

where $\mathbf{V} = \mathbf{P}/M$. The main influence of this effect is to modify the detuning parameter from $\Delta_0 = \omega - \omega_0$ to Δ , where $\Delta = \Delta_0 + \delta$.

If we assume a large atomic mass M , then we may ignore the recoil shift. The Doppler shift is a maximum for atoms at a given velocity travelling along the light beam direction and zero for those moving transverse to the beam direction. This is the reason for the well known crossed-beams configuration commonly adopted in experiments where an attempt is made to minimise the influence of the Doppler shift in controlling the detuning parameter.

We shall investigate the detailed nature of the shift given by Eq. (5) by considering a beam with azimuthal dependence. Many types of beam exhibiting azimuthal dependence, for example Laguerre-Gaussian laser modes or Bessel beams [8], may be shown to possess orbital angular momentum. We may note, however, that the beams need not be paraxial. Barnett and Allen [9] have shown that full beam solutions of the Maxwell equations which reduce to Laguerre-Gaussian beams in the paraxial limit, also possess intrinsic orbital angular momentum.

We consider the particular case of Laguerre-Gaussian beams which can be readily produced [4]. Such a beam is characterised by two integer indices l and p where l is the orbital angular momentum quantum number. Analytical expressions for the electric field vector of Laguerre-Gaussian modes in the paraxial approximation are well known [10]. The form that will be of concern to us here is essentially equivalent to light linearly polarised along the x -axis with an additional component of field along the direction of propagation. However, circularly polarised light may also be readily represented in the same formalism.

In order to highlight the angular momentum properties of the interaction of atoms with such Laguerre-Gaussian modes, it is sufficient to consider the special case of a dipole orientation along the x -axis such that $D_{12} = D\hat{x}$. More formally, the z -component of the electric field induces a component of the dipole moment in the z -direction; but this is almost everywhere in phase with the x -component and does not affect the result obtained in this special case. Only the x -component $\mathcal{E}_x$ of the electric field vector is involved in this case and we then have in cylindrical coordinates $\mathbf{R} = (r, \phi, z)$

$$f_0^l(\mathbf{R}) = ND\mathcal{E}_x(\mathbf{R}) = -i\omega NDF_l(r, z) \exp(-ikz) \exp(-il\phi), \quad (7)$$

where $F_l(r, z)$ contains no azimuthal dependence. The explicit form of F_l [11] is

$$F_l(r, z) = (-1)^p \frac{C}{w(z)} \left(\frac{r\sqrt{2}}{w(z)} \right)^l L_p^l(2r^2/w^2) \times \exp[-i(2p+l+1)\psi(z)] \times \exp[-i(kr^2/2\bar{z})] \exp[-(r^2/w^2)], \quad (8)$$

where L_p^l are associated Laguerre polynomials and we have set

$$\bar{z} = \frac{z_R^2 + z^2}{z}, \quad \frac{1}{2}kw^2(z) = \frac{z_R^2 + z^2}{z_R}, \quad \psi(z) = \tan^{-1}(z/z_R). \quad (9)$$

The quantities $\bar{z}$, $w(z)$ and $\psi(z)$ appearing in Eq. (8) are expressed in Eq. (9) in terms z_R , the Rayleigh range, which is defined by $z_R = \pi w_0^2/\lambda$ where w_0 is the beam waist. Thus f_0^l is, in general, a complex function of $\mathbf{R}$ and it is convenient to reformulate it as

$$f_0^l(\mathbf{R}) = G_l(\mathbf{R}) \exp[i\Theta_l(\mathbf{R})], \quad (10)$$

where $G_l(\mathbf{R})$ and $\Theta_l(\mathbf{R})$ are two real functions, easily deducible from Eqs. (7) and (8).

The corresponding shift arising from Eq. (5), denoted by δ_{LG} , is obtained as

$$\delta_{LG} = -i \frac{\nabla f_0^l \cdot \mathbf{V}}{f_0^l} - \frac{\hbar \nabla^2 f_0^l}{2M f_0^l}. \quad (11)$$

We shall assume that the atomic mass M is large and concentrate in what follows on the first term, ignoring

the recoil term. For convenience we continue to refer to δ_{LG} as the Doppler shift. We obtain, on substitution from Eq. (10),

$$\delta_{LG} = \nabla \Theta_l \cdot \mathbf{V} - i \frac{\nabla G_l \cdot \mathbf{V}}{G_l} = \delta_{LG}^{(1)} - i\delta_{LG}^{(2)}. \quad (12)$$

To leading order, the imaginary part $\delta_{LG}^{(2)}$ can be shown to be due to a change in the electric field amplitude arising from a change in the position of the atom from $\mathbf{R}$ to $\mathbf{R} + \mathbf{V}t$. The real part $\delta_{LG}^{(1)}$ can be expressed in cylindrical coordinates in terms of the components of the atomic velocity vector

$$\delta_{LG}^{(1)} = \nabla \Theta_l \cdot \mathbf{V} = \left[-k + \frac{kr^2}{2(z^2 + z_R^2)} \left(\frac{2z^2}{(z^2 + z_R^2)} - 1 \right) - \frac{(2p+l+1)z_R}{(z^2 + z_R^2)} \right] V_z - \left(\frac{kr}{\bar{z}} \right) V_r - \left(\frac{l}{r} \right) V_\phi, \quad (13)$$

where V_z , V_r and V_ϕ are the axial, radial and azimuthal velocity components of the atom. This is the main result of this article.

The axial Doppler shift is obtained from Eq. (13) as the first term (proportional to V_z). It is easy to see that this consists of the usual leading term $-kV_z$ plus terms accounting for the non-uniformity of the beam. The second contribution to the Doppler shift in Eq. (13) is the radial component and is given by the term proportional to V_r . This is again a property of the non-uniformity of the beam. A Hermite-Gaussian beam would also show such effects [12] and they are not the main concern of this report.

A much more significant effect resides in the final contribution. The corresponding azimuthal Doppler shift, denoted δ_{LG}^ϕ , is given by the V_ϕ term in Eq. (13),

$$\delta_{LG}^\phi = -\frac{V_\phi l}{r} = -l\omega_r. \quad (14)$$

The important features of this azimuthal shift are that it is directly proportional to the orbital angular momentum quantum number l of the Laguerre-Gaussian mode and that it is inversely proportional to the radial coordinates of the position of the atom.

A simple physical explanation of the azimuthal and radial Doppler shifts may be given in terms of the

Poynting vector. In a Laguerre-Gaussian mode the Poynting vector at any point is a tangent vector to a helix which winds around the beam axis [3]. At such a point the wave may be regarded as a local plane wave with its wavefronts normal to the Poynting vector. All the axial as well as non-axial components of the shift are directly related to the corresponding components of the local Poynting vector.

The main prediction of this article is that atoms with azimuthal velocity components relative to a Laguerre-Gaussian beam axis experience a frequency shift that is proportional to the orbital angular momentum quantum number l . The cylindrical symmetry of the cross-section of such beams would make them potentially attractive alternatives to the (0,0)-modes usually employed. Laguerre-Gaussian modes may readily be produced with either left- or right-handed orbital angular momentum for beams with l as big as 7 or more. It appears likely, therefore, that the azimuthal shift evaluated here may play an exploitable role in atom-field interactions, especially in experiments on the trapping and cooling of atoms and ions.

The effect may be particularly amenable to experimental measurement for the case of ions in a Penning trap, where, with an appropriate potential $U(R)$, the ions can be kept in a circular path of a fixed radius r with a constant velocity V_ϕ and undergo only minor perturbations in orbit due to interaction with the electromagnetic field of the light. In this very special case, the azimuthal shift can be given a transparent, albeit non-rigorous, interpretation. The relevant phase factor of the Laguerre-Gaussian mode $\exp(-il\phi)$ becomes $\exp[-il(\phi + V_\phi t/r)]$ after time t has elapsed, due to a change in the phase angle by an amount equal to the arc length $V_\phi t$ divided by the radius r . This phase factor immediately yields the azimuthal shift when written in the form $\exp(-il\phi - il\omega_r t)$ with $\omega_r = V_\phi/r$. In the case of atoms in a Penning trap the velocity V_ϕ is such that ω_r is typically of the order of a few MHz. The azimuthal shift is then several MHz, which is sufficiently large for experimental detection.

While an azimuthal shift could in principle be determined classically, we have chosen to use a quantum mechanical approach which proves to be very convenient. This has led to a quantum mechanical expression for all components of the shift in the resonant frequency arising from an arbitrary field distribution, including atom recoil. More importantly, perhaps, we

have established the basis of a theory for evaluating the interaction of the atom with Laguerre-Gaussian light beams where the explicit state of excitation of the system can be included by use of the optical Bloch equations.

The new component in the Doppler shift is just one of the manifestations of the orbital angular momentum effects of non-uniform beams. It is a potentially observable characteristic of the internal motion of the atom arising from its interaction with the light beam. Elsewhere we shall report on the angular momentum and torque due to such beams that are, by contrast, measurable properties of the gross motion of the atom.

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Rotational Frequency Shift of a Light Beam

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We explain the rotational frequency shift of a light beam in classical terms and measure it using a mm-wave source. The shift is equal to the total angular momentum per photon multiplied by the angular velocity between the source and observer. This is analogous to the translational Doppler shift, which is equal to the momentum per photon multiplied by the translational velocity. We show that the shifts due to the spin and orbital angular momentum components of the light beam act in an additive way. [S0031-9007(98)07769-2]

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The nonrelativistic, translational Doppler shift is a well-known phenomenon. The relative velocity v between light source and observer leads to a frequency shift $\Delta\omega = vk$, where $\hbar k$ is the linear momentum of each photon. The nonrelativistic rotational frequency shift identified in this paper can be described analogously by $\Delta\omega = \Omega J$, where Ω is the angular velocity between source and observer and $\hbar J$ is the total angular momentum per photon. We show that this result can be understood easily in terms of the rotational symmetry associated with the transverse electric field distribution of circularly polarized Laguerre-Gaussian beams.

It should be emphasized that the rotational frequency shift is completely different from the translational Doppler shift observed for a rotating source that arises from a linear velocity component in the plane of rotation (Fig. 1). The rotational shift is maximal when observed at right angles to the plane of rotation where the translational shift is zero.

It is well known that each photon in a light beam carries a spin angular momentum $\sigma\hbar$, where $\sigma = \pm 1$ corresponds to left- and right-handed circularly polarized light. Less well known is that light beams can also have an orbital angular momentum associated with the phase structure of the beam. In 1992 Allen *et al.* [1] showed for beams with an azimuthal phase term of $\exp(il\phi)$, such as Laguerre-Gaussian laser modes, that the orbital angular momentum is $l\hbar$ per photon. Laguerre-Gaussian modes have helical wave fronts with a phase discontinuity on the axis of the beam and an intensity distribution of one or more concentric rings [2]. These modes form a complete basis set; consequently, any arbitrary light beam can be described by a superposition of Laguerre-Gaussian modes.

For *spin* angular momentum alone, the rotational frequency shift has a number of precedents. In 1981 Garetz [3] attributed phenomena such as rotational Raman spectra and the frequency shift previously observed with a rotating half-wave plate [4] to an “angular Doppler effect.” In each case, a circularly polarized light beam rotating with an angular velocity Ω with respect to the observer produces a frequency shift of $\Omega\sigma$. Bialynicki-Birula and Bialynicka-Birula [5] predicted in a quantum mechanical calculation

that for an atomic system subject to a rotating potential a polarized transition experiences a frequency shift of ΩJ . We have shown recently [6] that the rotation of beams with an $\exp(il\phi)$ phase term, and so an *orbital* angular momentum of $l\hbar$ per photon, results in a frequency shift of Ωl . This shift can be related to previous predictions for the azimuthal Doppler shift of atomic transition frequencies [7] and to the frequency shift introduced by rotating cylindrical lenses [8]. In this paper we unify previous studies to include both spin and orbital angular momentum simultaneously. We find that the spin and orbital contributions behave in an additive way, such that it is the *total* angular momentum per photon, $\hbar J = \hbar(l + \sigma)$, that gives a rotational frequency shift of $\Delta\omega = \Omega J$.

In our experiment we rotate a circularly polarized beam possessing helical wave fronts and directly measure the corresponding frequency shift. The rotation of a source or detector without the introduction of unwanted off-axis

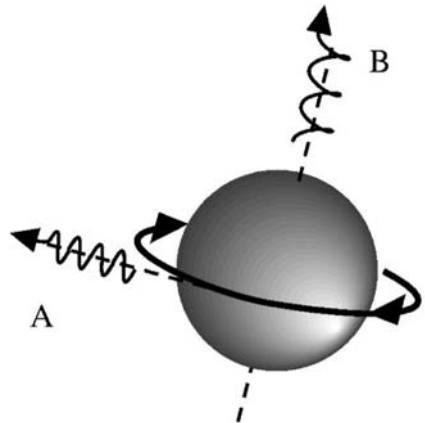


FIG. 1. Both the rotational frequency shift and the translational Doppler shift can be observed in the same rotating source. The frequency of light beam A is shifted due to the translational Doppler shift; light beam B is frequency shifted due to the rotational frequency shift.

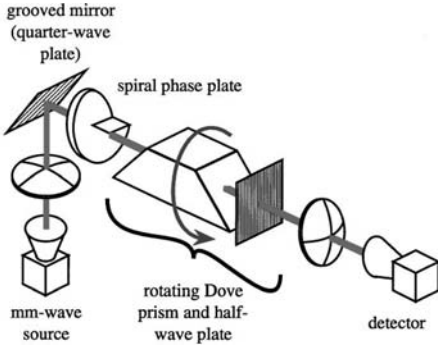


FIG. 2. Schematic diagram of the experimental apparatus for the observation of the rotational frequency shift.

movement is difficult to achieve. As an alternative, we use a Dove prism as an image rotator; the transmitted image and associated phase structure rotate at twice the angular velocity of the prism. Similarly, a half-wave plate rotates the electric field vector of a transmitted beam at twice the angular velocity of the wave plate. The combination of a Dove prism and a half-wave plate therefore rotates the whole beam while leaving the source stationary. To minimize difficulties in alignment of the prism with the beam axis, our experiment is performed in the mm-wave regime, using a Gunn diode source oscillating at 94 GHz with a corresponding wavelength of approximately 3 mm. Figure 2 shows the experimental arrangement. The source frequency is phase locked to a high harmonic of a crystal oscillator giving a short-term frequency stability of approximately 1 Hz. A corrugated feed horn couples approximately 10 mW into a linearly polarized fundamental Gaussian beam. This beam is collimated using a polyethylene lens and converted into a beam with helical wave fronts by transmission through a polyethylene phase plate whose thickness increases with azimuthal angle [9]. The beam is circularly polarized by reflection from a surface that has $\lambda/8$ deep grooves aligned at 45° to the incident linear polarization [10]. The frequency of the beam after transmission through the rotating polyethylene Dove prism and half-wave plate [10] is measured directly using a mm-wave frequency counter which is phase referenced to the source. The rotation of the beam introduces a shift in frequency. Reversing either the rotation direction or the sign of J changes the sense of the shift. Consequently, we expect the frequency difference between clockwise and counterclockwise rotation to be given by $\omega_{cw} - \omega_{ccw} = 2\Omega J = 2\Omega(l + \sigma)$. Figure 3 shows the observed frequency difference for various combinations of l and σ . The experimental observations agree with our predictions.

The rotational frequency shift could be interpreted in terms of an energy exchange with the rotating optical

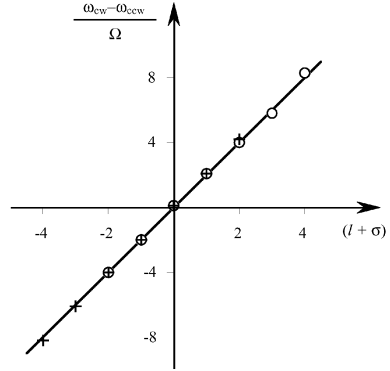


FIG. 3. Observed rotational frequency shift for various values of $(l + \sigma)$. The crosses and circles correspond to σ of -1 and $+1$, respectively. The beam angular velocity Ω was 1.25 Hz. The error associated with each measurement is 0.2 Hz.

element [8,11]. However, it is more illuminating to explain the shift in terms of the field distribution of circularly polarized Laguerre-Gaussian beams. As the electric field evolves in time, its distribution rotates. In each case the field distribution has $(l + \sigma)$ -fold rotational symmetry (Fig. 4) and the electric field undergoes a change of $l + \sigma$ cycles for each rotation of the beam.

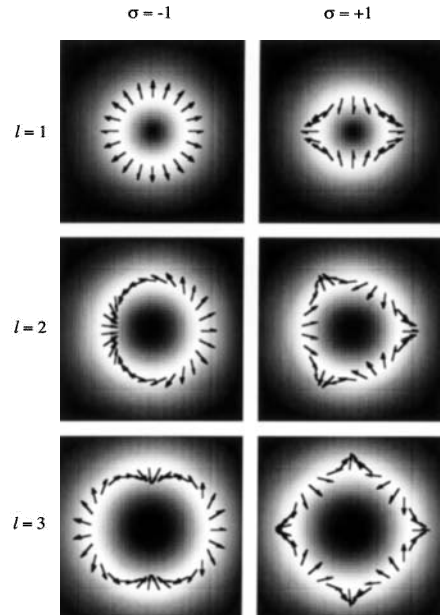


FIG. 4. Vector plots of the transverse electric field for circularly polarized Laguerre-Gaussian beams, showing the $(l + \sigma)$ -fold rotational symmetry.

For circularly polarized light the electric field rotates at the optical frequency, and therefore rotation of the beam results in a frequency shift. The direction of this shift depends on whether the sense of the beam rotation adds or subtracts to the rotation of the electric field. This is completely analogous to the translational Doppler shift which also depends upon a change in the number of cycles observed.

The translational Doppler shift is proportional both to the relative velocity and to the unshifted frequency. Although the rotational frequency shift is proportional to the angular velocity, it is independent of the unshifted frequency. This allowed us to perform our experiments in the mm-wave regime without reducing the magnitude of the shift. Unlike the translational Doppler shift, the rotational frequency shift depends on the distribution of the transverse electric field of the beam. Only for a pure circularly polarized Laguerre-Gaussian beam is there a single frequency shift and not a splitting. However, any arbitrary field distribution can be decomposed into a superposition of Laguerre-Gaussian beams; the resulting spectrum will consequently consist of sidebands at shifts proportional to the J value of each component.

We have described classically the rotational frequency shift, analogous to the translational Doppler shift, which arises from an angular velocity Ω between source and observer. For circularly polarized Laguerre-Gaussian beams the total angular momentum per photon is $\hbar J =$

$\hbar(l + \sigma)$. The resulting rotational frequency shift is given by $\Delta\omega = \Omega J$. Any field distribution can be expressed as a superposition of circularly polarized Laguerre-Gaussian beams and, if rotated, results in a frequency spectrum consisting of sidebands about the unshifted frequency. Specific combinations of Laguerre-Gaussian beams allow the generation of particular frequency distributions, which may have applications for ranging, communication systems or optical guiding of atoms.

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